

Acknowledgements

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A. Theoretical analysis

Appendix A.2 proves the main results under some assumptions about the kernel parameterization, using intermediate results about uniform convergence of our estimators in Appendix A.3. Appendix A.4 then shows that these assumptions hold for different settings of kernel learning.

A.1. Preliminaries

Given a kernel k_ω and sample sets $\{X_i\}_{i=1}^n \sim \mathbb{P}^n$, $\{Y_i\}_{i=1}^n \sim \mathbb{Q}^n$, define the $n \times n$ matrix

$$H_{ij}^{(\omega)} = k_\omega(X_i, X_j) + k_\omega(Y_i, Y_j) - k_\omega(X_i, Y_j) - k_\omega(X_j, Y_i);$$

we will often omit ω when it is clear from context. The U -statistic estimator of the squared MMD (2) is

$$\hat{\eta}_\omega = \frac{1}{n(n-1)} \sum_{i \neq j} H_{ij}.$$

The squared MMD is $\eta_\omega = \mathbb{E}[H_{12}]$. The variance of $\hat{\eta}_\omega$ is given by Lemma 10.

Lemma 10. *For a fixed kernel k_ω and random sample sets $\{X_i\}_{i=1}^n$, $\{Y_i\}_{i=1}^n$, we have*

$$\text{Var}[\hat{\eta}_\omega] = \frac{4(n-2)}{n(n-1)} \xi_1^{(\omega)} + \frac{2}{n(n-1)} \xi_2^{(\omega)} = \frac{4}{n} \xi_1^{(\omega)} + \frac{2\xi_2^{(\omega)} - 4\xi_1^{(\omega)}}{n(n-1)}, \quad (8)$$

where

$$\xi_1^{(\omega)} = \mathbb{E} \left[H_{12}^{(\omega)} H_{13}^{(\omega)} \right] - \mathbb{E} \left[H_{12}^{(\omega)} \right]^2, \quad \xi_2^{(\omega)} = \mathbb{E} \left[\left(H_{12}^{(\omega)} \right)^2 \right] - \mathbb{E} \left[H_{12}^{(\omega)} \right]^2.$$

Thus as $n \rightarrow \infty$,

$$n \text{Var}[\hat{\eta}_\omega] \rightarrow 4\xi_1^{(\omega)} =: \sigma_\omega^2.$$

Proof. Let U denote the pair (X, Y) , and $h_\omega(U, U') = k_\omega(X, X') + k_\omega(Y, Y') - k_\omega(X, Y') - k_\omega(X', Y)$, so that $H_{ij}^{(\omega)} = h_\omega(U_i, U_j)$. Via Lemma A in Section 5.2.1 of Serfling (1980), we know that (8) holds with

$$\begin{aligned} \xi_1^{(\omega)} &= \text{Var}_U [\mathbb{E}_{U'} [h_\omega(U, U')]] \\ &= \mathbb{E}_U [\mathbb{E}_{U'} [h_\omega(U, U')] \mathbb{E}_{U''} [h_\omega(U, U'')]] - \mathbb{E}_U [\mathbb{E}_{U'} [h_\omega(U, U')]]^2 \\ &= \mathbb{E}[H_{12}^{(\omega)} H_{13}^{(\omega)}] - \mathbb{E}[H_{12}^{(\omega)}]^2 \end{aligned}$$

and

$$\xi_2 = \text{Var}_{U, U'} [h_\omega(U, U')] = \mathbb{E} \left[\left(H_{12}^{(\omega)} \right)^2 \right] - \mathbb{E} \left[H_{12}^{(\omega)} \right]^2. \quad \square$$

We use a V -statistic estimator (5) for σ_ω^2 :

$$\hat{\sigma}_\omega^2 = 4 \left(\frac{1}{n} \sum_{i=1}^n \left(\frac{1}{n} \sum_{j=1}^n H_{ij}^{(\omega)} \right)^2 - \left(\frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n H_{ij}^{(\omega)} \right)^2 \right).$$

As a V -statistic, $\hat{\sigma}_\omega^2$ is biased. In fact, Sutherland et al. (2017) and Sutherland (2019) provide an unbiased estimator of $\text{Var}[\hat{\eta}_\omega]$ – including the terms of order $\frac{1}{n(n-1)}$. Although this estimator takes the same quadratic time to compute as (5), it contains many more terms, which are cumbersome both for implementation and for analysis. (5) is also marginally more convenient in that it is always at least nonnegative. As we show in Lemma 18, the amount of bias is negligible as n increases. In practice, we expect the difference to be unimportant – or the V -statistic may in fact be beneficial, since underestimating σ^2 harms the estimate of η/σ^2 more than overestimating it does.

Similarly, although we use the U -statistic estimator (2), it would be very similar to use the biased estimator $n^{-2} \sum_{i,j} H_{ij}$, or the minimum variance unbiased estimator $n^{-1}(n-1)^{-1} \sum_{i \neq j} (k(X_i, X_j) + k(Y_i, Y_j)) - 2n^{-2} \sum_{i,j} k(X_i, Y_j)$. Showing comparable concentration behavior to Proposition 15 is trivially different, and in fact it is also not difficult to show σ_ω^2 is the same for all three estimators (up to lower-order terms).